Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/13/2024
PI Name:	Weil, Michael Ph.D.		
Project Title:	Health Effects of Low Dose Rate Neutron Exposures		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.weil@colostate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	970-491-5902
Organization Name:	Colorado State University		
PI Address 1:	Department of Environmental & Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80521-2807	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2023	**End Date:**	12/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per S. Mack-Phillips/JSC (Ed., 6/10/25). NOTE: End date changed to 02/15/2025 per NSSC information (Ed., 11/7/2024).		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):	Brandl, Alexander Ph.D. (Colorado State University)		
Grant/Contract No.:	80NSSC23K1298		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Acute neutron exposures generally have high relative biological effectiveness (RBEs) for the induction of solid tumors, in the 10 to 70 range. Less is known about the carcinogenic effects of low dose rate neutron exposures. The limited rodent data available suggest either no dose rate effect, or low dose rate sparing, or an inverse dose rate effect depending on the tumor model investigated. The lowest published dose rate studied thus far is still 20-fold higher than what will be experienced by astronauts in deep space. Project 3 of "NSCOR: NASA Specialized Center of Research on Carcinogenesis" (the Carcinogenesis NSCOR investigation) was designed to collect carcinogenesis data on mice irradiated with fission spectrum, 252Cf neutrons at a dose rate of 1 mGy/day and with fission spectrum, reactor generated neutrons at an acute exposure dose rate. Total doses in the study are 0, 100, 200 and 400 mGy. Data from this study would increase our understanding of whether high cancer RBEs and inverse dose rate effects still occur at dose rates more relevant to space travel. This funding request is for support to monitor mice that received acute exposures. The funds will be used almost entirely for mouse maintenance per diems and animal care technician effort. In addition, animal care personnel currently supported on this project are approved users of the 252Cf irradiator/vivarium facility. They will be needed to run experiments for NASA funded investigators using that facility.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	The project aims to improve our understanding of carcinogenesis resulting from high LET radiation exposures in space. Similar exposures occur on Earth from Carbon ion therapy delivered to cancer patients and occupational exposures to some nuclear workers.
Task Progress:	The goals of this project were to irradiate samples for spaceflight-related studies in the low-dose rate neutron irradiator/vivarium facility at Colorado State University and to complete a sequencing study on spontaneous, gamma ray-induced, and HZE ion-induced hepatocellular carcinomas. The first of these goals was successfully accomplished with the irradiator facility being run 24 hours a day, 7 days a week during the reporting period. Samples were irradiated for NASA laboratories, the Italian Space Agency, and associated academic institutions and for a biotech company. It should be noted that this work is ongoing with requests for sample irradiations being received regularly. The second goal, completing the analyses of results in the hepatocellular carcinoma is nearly complete. The manuscript still requires revision and shepherding into publication.
Bibliography Type:	Description: (Last Updated: 04/27/2026)